

Routing Concepts — Guided Notes ANSWER KEY

Unit 1.8.1 | CompTIA Network+ N10-009 Obj. 1.8 | N10-008 1.8

For instructor use / distribute after students complete the notes.

Question 1

A router makes forwarding decisions by consulting its _____. Each entry contains the _____ network, the _____ IP or exit interface, and metadata including _____ and _____. If no match exists and there is no default route, the router _____.

Answer: Routing table; destination; next-hop; administrative distance and metric; drops the packet and sends ICMP destination unreachable.

Question 2

A routing table has entries for 10.0.0.0/8, 10.1.0.0/16, and 10.1.1.0/24. A packet destined for 10.1.1.75 will match _____ entries but be forwarded using the _____ route because of the _____ rule.

Answer: All three; 10.1.1.0/24 (/24); longest prefix match (most specific route wins).

Question 3

Connected routes appear automatically when _____. They require no manual configuration and have an administrative distance of _____. A static route has an AD of _____, meaning it is trusted _____ than a connected route.

Answer: An IP address is assigned to a router interface and the interface comes up; 0; 1; less (higher AD = less trusted).

Question 4

Static routing is appropriate for _____ networks because it is _____. It fails to scale because _____. Dynamic routing solves this by _____.

Answer: Small, stable; simple and predictable with no routing protocol overhead; every route must be manually configured and updated when topology changes; using protocols (OSPF, EIGRP, BGP) that let routers discover and share network information automatically.

Question 5

A default route uses the CIDR notation _____. It matches _____ destinations not covered by a more specific route. Home routers use a default route pointing to _____ so they don't need to know every route on the internet.

Answer: 0.0.0.0/0; all; the ISP's gateway.

Question 6

Fill in the default administrative distance values:

Connected route → _____

Static route → _____

OSPF → _____

EIGRP → _____

RIP → _____

Answer: Connected: 0; Static: 1; OSPF: 110; EIGRP: 90; RIP: 120.

Question 7

OSPF uses _____ as its metric, derived from _____. RIP uses _____ as its metric with a maximum of _____ hops. RIP is considered inferior because _____.

Answer: Cost; interface bandwidth (faster links = lower cost); hop count; 15; hop count ignores link speed — a path with fewer hops over slow links can win over a faster path with one extra hop.

Question 8 — Real World Application

REAL WORLD: A user reports they can reach some servers on the 10.2.x.x network but not others. You run 'show ip route' on the edge router and see a static route for 10.2.1.0/24 and an OSPF route for 10.2.0.0/16. A packet to 10.2.1.50 goes where, and why? A packet to 10.2.5.50 goes where, and why?

Answer: 10.2.1.50 → uses the static route 10.2.1.0/24 because it is more specific (longest prefix match) and also has lower AD (1 vs 110). 10.2.5.50 → uses the OSPF route 10.2.0.0/16 because the static route does not cover that address — only the /16 matches. If the static route's next hop is unreachable, 10.2.1.x hosts would be unreachable while 10.2.5.x hosts are fine, which matches the reported symptom of 'some servers reachable, others not.'